THE SHEAF CONDITION IS THE BIJECTIVITY OF THE MAP TO CECH H-0 FOR EVERY COVERING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument: it identifies the displayed map and states that the sheaf condition is exactly its bijectivity for every covering.

Editorial note. Cross-chapter restatement: the same label occurs in chapter cohomology for a topological space. That statement is the special case of this site statement, so per the lane-s4 rule only this more general occurrence is counted here.

Editorial note. The single labelled reference, which occurs inside the counted statement, points to a chapter of the Project that is not staged in this card's source tree; it is carried in assembly.external_references as a non-load-bearing navigation pointer recorded at the invoking line.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Cech complex and Cech H^0 of a covering. Same chapter, the Cech complex of a covering U and its degree zero cohomology as the equalizer of the two maps to the product over the fibre products. Supplies the right hand side of the displayed map.
- Sheaf condition for an abelian presheaf on a site. Same chapter, the definition of an abelian sheaf: for every covering, a section over U is determined by, and arises from, its restrictions to the members of the covering with the cocycle condition. Supplies the equivalence asserted, which is a restatement of that definition.
- The map from sections to Cech H^0. Same chapter, the natural map F(U) -> Cech H^0(U, F) sending a section to its restrictions. Supplies the map whose bijectivity is being characterised.

Source reference. Original section 00W1: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 862–862. Cross-chapter navigation pointer occurring inside the counted statement at source line 862, where the author refers the reader to the section on sheafification in the chapter Sites. The target chapter is not staged in this card's source tree, so the pointer records the invoking line; it only resolves the reference for the reader, and no bibliography citation is involved.

Lemma 0.1. *Let $\mathcal{C}$ be a site. Let $\mathcal{F}$ be an abelian presheaf on $\mathcal{C}$. The following are equivalent*

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(1) *$\mathcal{F}$ is an abelian sheaf on $\mathcal{C}$ and*
(2) *for every covering $\mathcal{U} = \{U_i \to U\}_{i \in I}$ of the site $\mathcal{C}$ the natural map*
$$\mathcal{F}(U) \to \check{H}^0(\mathcal{U}, \mathcal{F})$$
(see Sites, Section 00W1) is bijective.

Proof. This is true since the sheaf condition is exactly that $\mathcal{F}(U) \to \check{H}^0(\mathcal{U}, \mathcal{F})$ is bijective for every covering of $\mathcal{C}$. □